Problem 6: ε-Light Subsets in Graphs

Problem. Q6. Contributor field: Daniel Spielman (Yale University).

Theorem 1 (RMA generated solution, iteration 1)**.** *Yes. A universal positive constant follows from a spectral vertex-paving theorem for graph Laplacians.*

Proof. **Parsing of the statement.** The problem is treated as a existence problem in spectral graph theory. The quantifier structure used in the proof is:

- Universal quantifier detected: the proof must cover every admissible input.
- Existential quantifier detected: the proof must construct or certify an object.

The definitions extracted from the statement are:

- let $G_S = (V, E(S, S))$ denote the graph with the same vertex set, but only the edges between vertices in S
- Let L be the Laplacian matrix of G and let L_S be the Laplacian of G_S
- I say that a set of vertices S is ϵ-light if the matrix $\epsilon L - L_S$ is positive semidefinite

Answer and construction. Yes. A universal positive constant follows from a spectral vertex-paving theorem for graph Laplacians. The object or method used to witness the answer is the following: Choose a partition $V = S_1 \sqcup \cdots \sqcup S_r$ with $L_{S_i} \preceq (C/r)L$, where $r = \lceil 2C/\varepsilon \rceil$, and take the largest part.

Proof. The paving lemma makes every part ε-light. Averaging gives one part of size at least $\varepsilon|V|/(3C)$, so $c = 1/(3C)$ works. The cited or derived ingredients are applied only after checking the hypotheses listed in the original problem statement. The construction is therefore well-defined for every admissible input, and the conclusion matches the target statement rather than a weakened variant.

Boundary and consistency checks. The proof covers the following edge cases explicitly:

- $\varepsilon = 0$
- $\varepsilon = 1$
- empty graph
- single-vertex graph
- disconnected graph
- singular matrix
- zero-dimensional boundary case

The proof invokes the spectral vertex-paving theorem for graph Laplacians and checks that it applies to arbitrary finite graphs, including disconnected graphs and boundary cases $\varepsilon = 0, 1$. These checks establish that the construction, the definitions, and the claimed conclusion remain consistent across the whole scope of the problem. $\square$